SHUVANKAR DAS

MSc Mechatronics & Robotics | Leeds, United Kingdom
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PROFILE

Mechatronics and robotics engineer focused on medical devices, with a scholarship-funded MSc (Merit) from the University of Leeds. Designed rehabilitation exoskeletons with clinical stakeholder input, from CAD and FEA to EMG- and STM32-controlled prototypes, and an ISO 13485-aligned teleoperated surgical robot. Delivered a 46% bill-of-materials cost reduction and control latency under 100 ms. Seeking engineering roles in robotics, wearable robotics, medical technology, automation and embedded systems.

TECHNICAL SKILLS

Programming & Embedded: C, C++, Python, MATLAB, Bash/Linux, Git/GitHub, STM32, Arduino MEGA, ESP32, NodeMCU ESP8266, Arduino IDE, NVIDIA Jetson, RTOS concepts, CAN bus, FPGA/Verilog
Robotics, Vision & Sensing: ROS2, robot kinematics, control systems, EMG sensing, teleoperation, Pixhawk flight control, ArduPilot, soft robotics, OpenCV, Orbbec depth cameras, object detection pipelines, real-time vision processing, LiDAR, IMU, sensor fusion
CAD & CAE: SolidWorks, Catia V5 (advanced), Solid Edge, Siemens NX, PTC Creo, AutoCAD, Fusion 360, ANSYS FEA, CFD (Fluent)
Automation & Quality: Siemens PLC, SCADA, IoT/IIoT, MIG welding automation, Lean / Six Sigma, Poka-Yoke, root-cause analysis, GD&T, ISO 13485 (medical device design)

EDUCATION

MSc Mechatronics and Robotics (Merit) — Sep 2024 – Sep 2025
University of Leeds – Leeds, United Kingdom | Scholarship-funded | Comparable to NFQ Level 9

B.Tech Mechatronics Engineering (CGPA 7.9/10) — Aug 2020 – May 2024
Kalinga Institute of Industrial Technology (KIIT) – Bhubaneswar, India

EXPERIENCE

Product Research & Design Engineer (Part-time Internship, Remote) | Vain Automated Systems – Pune, India — Jan 2026 – Oct 2026

- Designed a wearable upper-limb exoskeleton with a 0–130° elbow range of motion, using a Feetech SM8524BL actuator and an EMG-sensor control system with 1 kHz sampling and under 100 ms latency.
- Reduced bill-of-materials cost by 46% (up to £1,000) through material selection and design simplification.
- Designed the frame in medical-grade nylon, keeping the complete structure under 1.4 kg, and applied GD&T to critical mounts and actuator housings.
- Validated the structure in ANSYS 2024 FEA for users up to 130 kg, identifying failure modes under dynamic loading.
- Iterated the design with clinical stakeholders, using their therapeutic feedback to improve usability.

Research Engineer (Internship) | Indian Institute of Technology Kanpur (YRRNA Labs) – Kanpur, India — Jun 2023 – Jul 2023

- Carried out a comparative study of military tanks worldwide to set the baseline for a new tank design proposal.
- Researched existing tank powertrains and designed a powertrain and multi-fuel engine concept for the proposed vehicle.
- Researched automated and remotely operated systems to reduce the personnel needed on the battlefield.

Industrial Trainee, Assembly Automation | Maruti Suzuki India Limited – Gurugram, India — Jun 2022 – Jul 2022

- Programmed and monitored PLC logic and SCADA dashboards for process control on high-volume assembly lines, gaining exposure to Siemens ladder logic, HMI configuration and sensor feedback loops.
- Designed and simulated fixture and tooling layouts in Solid Edge to reduce cycle time and equipment idle time.
- Assisted with integration and troubleshooting of robot arms, vision inspection systems and pneumatic actuators, supporting line uptime and OEE.
- Took part in Lean and Poka-Yoke error-proofing work, applying root-cause analysis to reduce defects and improve first-pass yield.

PROJECTS

Upper-Limb Exoskeleton for Stroke Rehabilitation | University of Leeds — MSc Dissertation

- Designed in SolidWorks and built a prototype under 3 kg with STM32 control and selectable rehab programmes, verified by FEA.

Autonomous UAV for Waterway Waste Removal | University of Leeds — Sep 2024 – Nov 2024

- Led design of a 20 kg UAV with a 360° robotic arm, ROS2 autonomy, Pixhawk flight control and OpenCV depth vision.

Medical Teleoperation Surgical Robot | University of Leeds — Sep 2024 – Nov 2024

- Designed a teleoperated 6-DOF robot for knee arthroplasty to ISO 13485, with Maxon motors and storage for 20+ surgical tools.

Ankle Rehabilitation Exoskeleton | University of Leeds — Sep 2024 – Dec 2024

- Led design of a tendon-driven 3-DOF exoskeleton over 4+ iterations and 30+ tests; reduced clinical visit frequency by 30%.

Autonomous Linear Welding Robot | KIIT — Dec 2022 – Apr 2023

- Built an Arduino MEGA MIG welding robot with safety interlocks and ESP8266/Blynk IIoT monitoring; ~50% less operator time.

IC Engine Go-Kart | Formula Student (SAE India) — Jan 2021 – May 2024

- Led design of a 127 cc kart: 33 kg chassis (SolidWorks, ANSYS FEA/CFD), CVT integration and ESP32 data acquisition.

CERTIFICATIONS

SolidWorks CSWA • Developing Industrial IoT (Coursera) • Modelling & Debugging Embedded Systems • Introduction to Programming with MATLAB • Introduction to Self-Driving Cars (University of Toronto)